

# Chapter I

## Introduction to R for Statistics

### 1 Basic Syntax

|                                        |                                                                         |
|----------------------------------------|-------------------------------------------------------------------------|
| <code>data &lt;- "data/"</code>        | Initialisation of a string object                                       |
| <code>getcwd()</code>                  | Function for getting the current working directory                      |
| <code>setcwd(dir)</code>               | Function for changing the current working directory to <code>dir</code> |
| <code>paste(path, file, sep="")</code> | Combine every string into a filepath with no separation                 |

---

```
1 face_data <- read.csv("data-files/face-data.csv", sep=",", header=TRUE, dec=".",  
  ↪ stringsAsFactors = TRUE)
```

---

`face_data` is a `data.frame` object in R, `read.csv()` is a function that read a `.csv` file and return a `data.frame`. `stringsAsFactors` is just to make the string values to be treated as categories, for example if there are:

---

```
1 "Junior", "Sophomore", "Senior"
```

---

Then there is 3 categories, for instance, for a column of `Undergrad_level` whose value per row is either of those. `sep=","` is just the separator between column is the comma, `header=TRUE` means first line contains variable names (headers). The `dec="."` means that the starting decimal of numbers is behind “.”.

|                                                    |                                                                                 |
|----------------------------------------------------|---------------------------------------------------------------------------------|
| <code>head(data)</code>                            | If not specify number of line it will print the first 6 lines and the headlines |
| <code>tail(data, 10L)</code>                       | Print the last 10L lines                                                        |
| <code>summary(data)</code>                         | Provide a description of summary of statistic of the data.                      |
| <code>summary(face_data\$rating)</code>            | Using the <code>\$</code> to specify which headline column to use.              |
| <code>summary(as.factor(face_data\$rating))</code> | <code>as.factor</code> is used for change the numerical quantity to categories. |
| <code>dim(data)</code>                             | This is for getting the dimension of the data frame.                            |

## 2 Matrix, Vector and Scalar

In R there are many objects type firstly

---

```
1 id <- 10
```

---

An object called `id` as a variable valued 10. As of now, the `read.csv()` function return a `data.frame` which is an extension of another object called *matrix*, that is, a collection of numbers ordered by rows and columns created by `matrix()` function. Therefore, `data.frame` is a similar object as matrix but can contain multiple data type such as both strings and numerical values.

To create a matrix,

---

```
1 M <- matrix(c(1:9), nrow=3)
```

---

To index the matrix,

---

```
1 M[i,j]  # Row and column addressing
2 M[i,]   # Row slicing
3 M[,j]   # Column slicing
```

---

Either each row or column of a matrix is a *vector*, always used by R under the hood, that is, there is always a `[1]` in front of the printed value. Each single entry of the vector is a *scalar*. The function `c()` can be used to create vector, for example, `c(1:9)` or `1:9` would suffice to create a vector.

To concatenate a vector,

---

```
1 id <- 10
2 id <- c(id, 11)
```

---

To index the `data.frame`, it inherits the indexing mechanism from matrix with the addition of

---

```
1 df$<wanted_column>[i]
2 df$<wanted_column>[i:j]
3 df$<wanted_column>[df$<filter_column><condition>].
4 # For example
5 face_data$gender[face_data$rating>95]
```

---

## 3 Measurement Levels

1. *Nominal*: Nominal data makes a distinction between groups (or sometimes even individuals), but there is no logical order to these groups. For example, in voting, the voters are Democrats or Republicans but there is no measurement between voters of how much better, how much smarter. In R, nominal data is often encoded as `factor` or forced by `as.factor()` which are then a *variable* recall that it is the *column* of the `data.frame`. The values that the variable can take are stored separately by R and are referred to as the different *levels* of the factor.
2. *Ordinal*: Ordinal data makes distinction between groups or individuals but now imposes an order. For example, Gold medal, Silver medal, and Bronze medal in a contest.
3. *Interval*: Interval data distinguishes group by imposing an order, and measuring the difference in some unit. For example, the HD students scores 95, the D best scores 90, and the C scores 70. The order is rendered by the difference of scores between students. The interval is the score, for example, `[91, 95]` are the “interval” for HD students. Since there is not starting point, in other words, an initial “0” so I cannot say that HD student is 91/70 times better than the C students.

4. *Ratio*: Ratio data contain all the attributes of Interval data but have a clear reference point or “0”. For example, if the lowest score is 0 then measuring from the lowest to highest we know that the ratio between the smartest and worst is  $95/70 \approx 1.35$ .

Each higher measurement level contain as much information as the lower and more. Therefore, the higher data type can be summarise into the lower level data type. *Nominal and Ordinal* data are often called *categorical data*, while *Interval and Ratio* are often called *numercial data* comprising *continuous and discrete* data.

## 4 Outliers and Unrealistic Values

Sometimes invalid data appear and it need to be removed for example the invalid "" inside the gender column. I can remove the rows of of data which has the invalid gender by the code,

---

```
1 face_data <- face_data[face_data$gender == "Female" | face_data$gender == "Male", ]
```

---

Note that the code is indexing by row splicing given either Male or Female. After the removal of the rows containing the empty value, R still consider empty value as a possible *level* of the factor of gender, to drop this level explicitly,

---

```
1 face_data$gender <- droplevels(face_data$gender)
```

---

In statistics, there are *plausible* and *implausible* outliers, that is, a 600-year-old person is implausible but a 500kg person is plausible. To deal with implausible outliers, for simplicity of now, there are

1. Ignoring them
2. Removing them
3. Substituting them using statistical method (Advanced)

Removing implausible outliers is a common practice but should it be prohibited for plausible outliers.

For missing values either those are ignored or removed, or filled with value created by prediction model called imputation. Note that

Researchers also like to use special values to indicate missing data, like 88, 888, 99, and 999. These values should be used only when they represent implausible values for the variable. Thus 99 for a missing age is not recommended, while 999 would be proper value for a missing age.

## 5 Describing Data

### 5.1 Frequency

---

Nominal and ordinal data are often described using frequency tables. In the frequency counting there are frequency of occurence, cumulative frequency, relative frequency, and relative cumulative frequency.

#### Definition

---

Given a set of number (as indication of each level), that is,  $\{x_1, x_2, \dots, x_n\}$  where  $n$  is the last level. For example,  $x_1 = 1$  is ages between 18 to 24. The frequency

for each  $x_i$  is  $f_i$  for  $i = 1, 2, \dots, n$ . The cumulative frequency for  $x_j$  is,

$$\sum_{k=1}^j f_k$$

The relative frequency of  $x_j$  is,

$$\frac{f_j}{\sum_{k=1}^n f_k}$$

The cumulative relative frequency is,

$$\frac{\sum_{k=1}^j f_k}{\sum_{m=1}^n f_m}$$

`transform(data, ...)` function in R is used to transform the data, the first argument is the data to be transformed. In the next argument position, those are new columns to be added, that is, variable name and operation to assign value to the variable. The `table()` function will create a table of categories and its frequency count. As of now, the table has two objects which are the category and the `Freq` used in the `transform()`. `prop.table` or `proportions()` give a percent of each count divided by the total, that is,  $f_i / \sum f$ , the relative frequency. For example, to create table data that has frequency count, cumulative frequency, and relative frequency.

---

```
1 t1 <- table(face_data$age) ##
2 t2 <- transform(t1, cumulative = cumsum(Freq), relative = prop.table(Freq))
```

---

Frequencies are often uninformative for interval and ratio. If there are numerous categories, each category will have few counts, for instance, count of 1, resolved by *discretising* or, in other words, *binning* which “effectively discards some of the information in the data”. To discretising the variable,

---

```
1 bins <- <an_integer>
2
3 # As of now, the two rating_binned seems to be identical.
4 rating_binned_factored <- factor(cut(face_data$rating, breaks=bins))
5 rating_binned <- cut(face_data$rating, breaks=bins)
6
7 t1 <- table(rating_binned)
8 t2 <- transform(t1, cumulative = cumsum(Freq), relative = proportions(Freq))
```

---



---

```
1 rating_binned Freq cumulative relative
2 1 (0.901,20.8] 381 381 0.1054234
3 2 (20.8,40.6] 513 894 0.1419480
4 3 (40.6,60.4] 821 1715 0.2271721
5 4 (60.4,80.2] 1214 2929 0.3359159
6 5 (80.2,100] 685 3614 0.1895407
```

---

## 5.2 Central Tendency

Each numerical variable exhibits a “central value” or “middle value” referred to as the location of the data. Some prerequisites of R,

1. `unique()`: returns a vector of all unique values.

2. `which(<logical_object>)`: returns indices of the logical object.
3. `which.max(<object>)`: return the index of the the first largest value of the object.
4. `tabulate()`: return a table which uses the object's values as positions of the table while the second column is about the object's frequency of occurrence.
5. `match()`: "return vector of the positions of first matches of its first argument in its second".

**Arithmetic Mean** Arithmetic mean,  $\bar{x}$ , "is a sample mean of a variable" affected by extreme outliers because such mean weighs every value equally given by the formula,

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad (\text{I.1})$$

where  $n$  is the total samples,  $x_i$  is the value comprised by the variable. Use `mean()`.

**Mode** Mode is the most frequently occurring value of a variable and said to have multiple modes in a variable. In R,

---

```
1 get_mode <- function(v){
2   uniqv <- unique(v)
3   uniqv[which.max(tabulate(match(v, uniqv)))]
4 }
5 get_mode(face_data$rating)  ##
```

---

**Median** Being not necessarily in the data, median divides the *ordered data* in two equal parts, that is, 50% less than the median and 50% otherwise. When  $n$ , amount of data, is odd, median is the middle value. When  $n$  is even, median is the *arithmetic mean* two middle values. In practice, sort the data first then find the median. Use `median()` in R.

**Quartiles, deciles, percentiles, quantiles** Being not necessarily in the data, QDCQ of an *ordered data* are represented by *cut-off* values at certain  $i\%$  where 0% is the minimum value, and 100% is the maximum. More formally,

A *quantile*  $x_q$ , where  $q \in [0, 1]$ , divides the data in two parts, that is,  $q \cdot 100\%$  of data is below  $x_q$ , and  $(1 - q) \cdot 100\%$  of data is otherwise.

1. Quartile:  $q \in \{0.25, 0.5, 0.75\}$  consecutively referring to first, second, and so on quartile. The second quartile is a median. Note that 0% and 100% are not included.
2. Decile:  $q \in \{0.1, 0.2, \dots, 0.9\}$
3. Percentile:  $q \in \{0.01, 0.02, \dots, 0.99\}$

"Thus quartiles and deciles are also percentiles." Use `quantile()` in R.

The quantile's values vary depending on the our interpolation of *distance between two values*. Practically, map any data onto the  $[0, 1]$  interval (such action is called *normalisation*?, I assume). Generically, given by,

$$q = \frac{i}{n + 1} \quad (\text{I.2})$$

where  $i$  is the  $i^{th}$ -position value,  $n$  is the amount of data,  $q \in [0, 1]$ . However, the default setting in R,

$$q = \frac{i - 1}{n - 1} \quad (\text{I.3})$$

For instance using the generic method with the given the sorted data set  $\{2, 4, 5, 6\}$ , I have that  $q_2 = 1/5$  and  $q_4 = 2/5$ . For the 0.25 quartile, it should be positioned at  $q = 0.25$  being the value 2.5 which is the first quartile. However, if use the R's method, the second quartile will be 3.5. Note that `quantile(data, type=<type_of_interpolation>)`.

### 5.3 Dispersion, Skewness, and Kurtosis

Dispersion indicates distance from the centre location or each other while skewness and kurtosis “are measures describing the shape of the frequency plot”. Note that throughout those operations, there emerges  $x_i - \bar{x}$  which is trying to quantifies the distance of data from the mean. Noticing such with an understanding helps to discover a glimpse into the variance, std, the skewness and kurtosis, etc.

Regarding skewness, instead of thinking of the plot of how skewness works, using this intuition, by my assumption, I can also visualise how data are comparing to each other, since it is right-skewed, this mean that by the formula, distance of value on the right is further away from the mean and far away from each other as well, that is why the tail on the right is longer as in how dense the plot is.

**Range and interquartile range** Range is the “difference between the maximum and minimum” while interquartile range, IQR, “calculates the difference between the third quartile and the first quartile”. Differing from *range* being sensitive to extreme outliers, IQR circumvents such problem by quantifying a range of 50% of data. In R,

---

```
1 quantile(face_data$rating, c(0.75)) - quantile(face_data$rating, c(0.25))
```

---

**Mean absolute deviation** Mean absolute deviation computes the average distance of data being from the mean, given by,

$$\text{MAD} = \frac{1}{n} \sum_{i=1}^n |x_i - \bar{x}| \quad (\text{I.4})$$

In R,

---

```
1 sum(abs(x-mean(x)))/length(x)
```

---

**Mean squared deviation, variance, and standard deviation** Similar to MAD, however MSD uses square root, that is,

$$\text{MSD} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \quad (\text{I.5})$$

Variance differs from MSD when sample size is small but almost identical for large sample. Explained by the formula,

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1} = \frac{n \cdot \text{MSD}}{n-1} \quad (\text{I.6})$$

Standard deviation (std) “is on the same scale as the original variable, instead of a squared scale for the variance.”

$$s = \sqrt{s^2} \quad (\text{I.7})$$

**Skewness and kurtosis** Skewness and kurtosis both are computed by a *standardised* value — *z*-value — having no unit, that is,

$$z_i = \frac{x_i - \bar{x}}{s} \quad (\text{I.8})$$

Therefore, it means and variance are consecutively 0 and 1, given by,

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n z_i &= \frac{1}{n} \cdot \frac{\sum_{i=1}^n (x_i - \bar{x})}{s} = 0 \\ \frac{1}{n-1} \sum_{i=1}^n (z_i - 0)^2 &= \frac{1}{n-1} \cdot \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{s^2} = s^2/s^2 = 1 \end{aligned} \quad (\text{I.9})$$

Skewness –  $g_1$  – measures the asymmetry of a plot while kurtosis –  $g_2 \in [-0.5, 1.5]$  – measures the peakedness of the plot. Both  $g_1$  and  $g_2$  can be either positive or negative, and unchanged when all values shifted by a fixed number or when they are multiplied with a fixed number, in other words, such operations on data do not change the shape.

When  $g_1$  is positive or called right-skewed, the values on the right side of the mean are further away from each other and the mean. Thus the tail on the right is thinly long. When  $g_1$  is negative or called left-skewed, the same logic applied. When  $g_1 = 0$  data is symmetric, or  $|g_1| \leq 0.3$  is near to symmetric.

When  $g_2$  is positive, it is called *leptokurtic* having “long heavy tails and is severely peaked in the middle”. When  $g_2$  is negative, it is called *platykurtic* having “tails of the data are shorter with a flat peak in the middle”. When  $g_2 = 0$ , the data is called *mesokurtic*.

The two formulae,

$$g_1 = \frac{1}{n} \sum_{i=1}^n \left( \frac{x_i - \bar{x}}{\sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}} \right)^3 = \frac{1}{n} \sum_{i=1}^n \left( \frac{x_i - \bar{x}}{s} \right)^3 \quad (\text{I.10})$$

$$g_2 = \frac{1}{n} \sum_{i=1}^n \left( \frac{x_i - \bar{x}}{\sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}} \right)^4 - 3 = \frac{1}{n} \sum_{i=1}^n \left( \frac{x_i - \bar{x}}{s} \right)^4 - 3 \quad (\text{I.11})$$

## 5.4 A Note on Aggregated Data

Practically, sometimes data has been summarised such as in groups or intervals, for example, counting people in salary group. To find the descriptive statistics of  $n$  groups, the median  $x_j$  of each group  $j$  must firstly be calculated and with the frequency  $f_j$  giving the formulae,

$$\begin{aligned} \bar{x} &= \frac{\sum_{k=1}^n x_k f_k}{\sum_{i=1}^n f_i} \\ s^2 &= \frac{\sum_{k=1}^n (x_k - \bar{x})^2 f_k}{\sum_{i=1}^n f_i - 1} \end{aligned} \quad (\text{I.12})$$

## 6 Visualising Data

### 6.1 Describing Nominal/Ordinal Variables

Frequency is useful for describing Nominal and Ordinal data using *bar chart* to emphasise on absolute numbers of frequencies and to compare the height (counts) of each value (or group), while using *pie chart* to emphasise on the relative frequency and to indicate the share of each value. In R,

---

```
1 counts <- table(face_data$age)  ##
2 t1 <- transform(counts, relative=proportions(Freq))
3 pie(t1$relative)
4 barplot(t1$Freq)
```

---

There are more methods, as well as the advanced ones, but bar and pie chart would suffice for now.

---

## 6.2 Describing Interval/Ratio Variables

---

**Box Plot** Here shows the method to visualise continuous variables. Firstly the *box plot*, in other words, *box and whiskers* plot which is “useful for getting a feel of the spread, central tendency, and variability”.

Middle bar is the median, top of the box is  $Q_3$ , whereas bottom of the box is  $Q_1$ . The whiskers are the smallest value  $m$  and the largest value  $\ell$  being

$$\begin{aligned} m &\geq Q_1 - 1.5\text{IQR} \\ Q_3 + 1.5\text{IQR} &\geq \ell \end{aligned} \tag{I.13}$$

As I assume that  $m$  and  $\ell$  cannot be thought of as the min and max since those could be outliers which are the dots being

$$x_i \notin [Q_1 - 1.5\text{IQR}, Q_3 + 1.5\text{IQR}] \tag{I.14}$$

In R,

---

```
1 boxplot(face_data$rating)    ##
```

---

**Histogram** Histogram, a continuous variant of bar chart, discretising (Section 5.1) the data, then shows the frequency in each bin. Note that “too few bins obscure the patterns in the data, but too many bins lead to counts of exactly one for each value”. Use `breaks` to change the bins,

---

```
1 hist(face_data$rating)
2 hist(face_data$rating, breaks=5)
3 hist(face_data$rating, breaks=50)    ##
```

---

**Freedman-Diaconis Rule** One of the ways to determine the number of bins  $b$  is Freedman-Diaconis rule, where the bin width  $w$  is given by,

$$\begin{aligned} w &= 2n^{-1/3}\text{IQR} \\ b &= \frac{\max - \min}{w} \end{aligned} \tag{I.15}$$

**Density Plot** As of now, density plot can be understood as “a continuous approximation of a histogram” yielding a probability per range of values. The higher the line the more likely are the values to fall in that range. In R,

---

```
1 plot(density(face_data$rating))    ##
```

---

---

## 6.3 Relations Between Variables and Multi-panel Plots

---

*Scatter plot* is used for describing the relationship between two continuous variables. Each dot of the plot is a unique  $(x, y)$  pair. *Box plot* is used to show the relationship between a categorical variable and a continuous variable. In R,

---

```
1 plot(face_data$dim1, face_data$rating)    # R automatically produce a scatter plot
  ↪ given two cont. vars
2 plot(face_data$gender, face_data$rating)    # R automatically produce a box plot
  ↪ given 1 cat. var and 1 cont. var
```

---



In R, use `par(mfrow = c(x, y))` to set up a *canvas* for plotting, *x*, *y* are consecutively the row and column size for how many plots on the canvas. For example for 6 graphs in R,

---

```
1 par(mfrow=c(3,2))
2 plot(face_data$gender)
3 plot(face_data$gender, face_data$rating)
4 pie(counts)
5 barplot(counts)
6 hist(face_data$dim1)
7 plot(density(face_data$dim2))  # $
```

---

## 6.4 Plotting Mathematical Functions

---

To plot a math function, traditionally, firstly define a **function** in R with the argument *x*. Then, use `seq()` to generate a sequence of number, check `?seq` for help manual. Finally, run `plot()`, use `type="l"` for a line plot instead of the default dots, or `type="b"` for both line and dots. Alternatively, use `curve(<function_name>, xlim=c(<start>, <end>))` to plot but make sure the function accept an *x* argument, and *xlim* is the domain.

**Frequently Used Arguments** R's simple `plot()` function can be tailored to our needs. Using the following

1. *type*: type of plot
2. *xlim*, *ylim*: bound of either *x*-axis or *y*-axis using `c(<upper>, <lower>)`
3. *lwd*: set line width
4. *col*: set colour of the plot (not whole graph), can be used with `c(<number>, <number>)`
5. *xlab*, *ylab*: Labelling
6. *main*: the top caption

## 7 Problem

7.1 1

---

```
1 get_modes <- function(v){
2   uniqv <- unique(v)
3   freq_oc_table <- tabulate(match(v, uniqv))
4   max_oc <- max(freq_oc_table)
5   max_oc_pos <- which(freq_oc_table==max_oc)
6   modes <- uniqv[max_oc_pos]
7   # return(unique(modes))
8   return(modes)
9 }
```

---

7.2 2

In R use,

---

```
1 dat <- dat[complete.cases(dat) & dat$Voting != 99 & dat$Age !=622,]
```

---

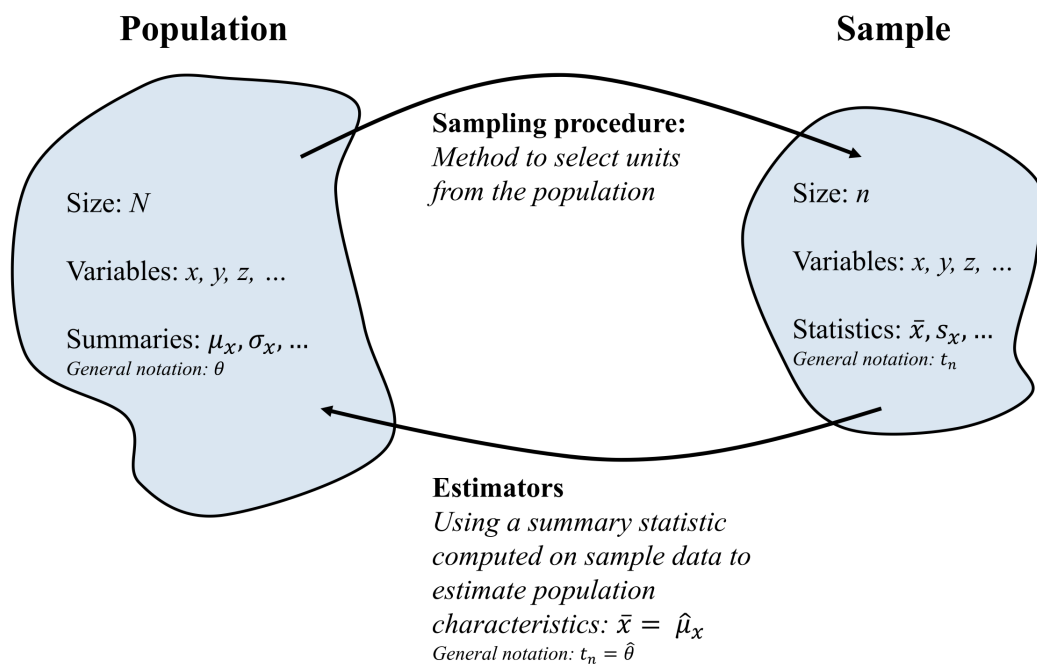
where `complete.cases(<data.frame>)` is to spot the missing value or NA, which returns a boolean vector.

# Chapter II

## Sampling Plans and Estimates

### 1 Introduction

*Statistical inference* is “to extend the conclusion beyond the observed data”. The origin or source of the data is important. For example, in the university, a selection both all female and male from different courses, but the selection only based on the GPA on the higher spectrum. Then the “dataset is considered *selective* towards” high-scoring students. Those issues can be overcome by the understanding of population, sample, sampling procedures, and estimation of population characteristics



A dataset contains multiple units. In the case of the units of the sample data comprising  $\{x_1, x_2, \dots, x_n\}$  which is a subset, not necessarily the same index, of the population,  $\{x_1, x_2, \dots, x_N\}$ . To create a sample, I do *sampling procedures*, formal probabilistic approaches, whose values “are referred to as a realization from the population”.

Knowing the sampling procedures, I know how close the *estimates*, calculation on sample data, to the *theoretical value*, “a calculation procedure applied to all units in the population”, in the population and also have precise statistical inference. Note that it is called “theoretical” value because each unit may not be in population but in our belief.

## 2 Definitions and Standard Terminology (cont.)

A *unit* “is usually a concrete or physical thing for which we would like to measure its characteristics”. For example, unit as human in medical, social science, etc. or as images in photography, computer vision. The sets of units (SOU) to obtain data is a sample which is a subset of a complete SOU, the (target) population. Note that, sometimes there exists units not from the target.

“Statistical inference is the process of deducing properties of an underlying population by analysis of the sample data. Statistical inference includes testing hypotheses for the population and deriving population estimates.”

Intuitively, as of now, units can be thought of as the *id* in many dataset given that each of that corresponds to a unique thing (unit).

A *unit* can be *sample unit* or *observation unit*. Based on my assumed intuition, for example, each sample has set of *id*’s (sample units), each *id* contains multiple “hidden” attributes (the columns next to it) depending on what to estimate, those are the observation unit. Say, estimating university’s output quality, the sample unit is the student whose observation units are degree, achievement, GPA, etc.

Defining a population is hard since there are so many ambiguities to be unveiled, and sampling the whole population is impractical. Thus, we rely on the a small sample which can be from *non-representative sampling* or *representative sampling*. The first one implies that the exact process of getting units is unknown. The second one implies the units approximate probability distribution of the population and therefore referred to as *random* or *probability sampling*.

## 3 Non-representative Sampling

“It is strongly recommended not to apply these methods, unless knowledge is available on how to adjust or correct the sample for inferential purposes.”

---

### 3.1 Convenience Sampling

Convenience sampling collect data that is easily attainable, which only depicts a small fraction of the whole population, provides a *biased sample*. Such sampling “is often justified by using the argument of population homogeneity”, with such assumption it is irrelevant to sample anymore since all of population is the same.

---

### 3.2 Haphazard Sampling

Sample something by random but have a tendency to select something based on our unconscious preference.

---

### 3.3 Purposive Sampling

Purposive sampling or judgmental sampling tries to sample units for a specific purpose. Such sampling is useful to infer specific knowledge from the *population of interest*, but not general statements about the whole population. For example, I am interested about the electricity usage during winter, so I only include the data of months in the winter period.

## 4 Representative Sampling

In NRS, some units are more likely to appear in data but importantly its quantity is unknown. In RS, aka a control randomness using probability, the likelihood of a specific unit occur in the sample is known. For example, a rolling dice has 6 surfaces, each can appear with probability of  $1/6$ . Therefore if a unit to be shown with a probability  $1/m$  then create  $m$  sides. Practically, use pseudorandom numbers to select unit.

## 4.1 Simple Random Sampling

SRS is a way to collect units from the population with the same probability, but it is proved to be hard in practice. However, it is important in statistical theory. In R,

---

```
1 x <- c(1:20)
2 set.seed(575757)
3 # FALSE means sample without replacement aka the Combination
4 s_dat <- sample(x, 5, replace=FALSE, prob=NULL)
```

---

For example, let  $N = 6$ , and  $n = 3$  (sample size), then there are  $K = 20$  (combinations). Therefore, the probability of each combination is  $p_c = 1/20$ , but  $p_u = 1/2$  for each unit. By the general property of SRS,

“Each unique sample (or combination) has the same probability of being selected, and, as a result, each unit has the same probability of being selected.”

Each unit in the population appears 10 out of 20 times in the combination, if listing out all the combination.

It is cumbersome to calculate the total number of samples or combinations then the probability, lastly select the random one. The combination of  $K$  is given by,

$$K = \frac{N!}{n!(N-n)!} \quad (\text{II.1})$$

As  $N$  gets larger the complexity increase tremendously. Similarly for  $n$ .

**Sequential Sampling** A practical approach to SRS is that picking each unit with a probability to form a sample. With the same example, let  $S_j = \{x, y, z\}$  be a sample where  $i \in \{1, 2, \dots, 20\}$ , and  $x, y, z$  are the units. Sequentially picking a unit *without replacement* would gives the following probability for each unit,

$$\begin{aligned} p_i &= 1/6 \\ p_j &= 1/5 \\ p_k &= 1/4 \\ p_i \times p_j \times p_k &= 1/120 \neq 1/20 \end{aligned}$$

However, there are  $3!$  variants (duplicate), hence  $p_{S_i} = 3! \times 1/120 = 1/20$

### SRS's Property

“Each unique sample (or combination) has the same probability of being selected, and, as a result, each unit has the same probability of being selected.”

An informal proof can be demonstrate as the following,

*proof*

For the first unit  $i = 1$ , the combination of the population without the first unit is

$$\frac{(N-1)!}{n!(N-1-n)!}$$

Hence the probability of  $i = 1$  not in samples is,

$$\frac{\frac{(N-1)!}{n!(N-1-n)!}}{\frac{N!}{n!(N-n)!}} = \frac{\cancel{(N-1)!} \cancel{n!} \cancel{(N-n)!}^{N-n}}{\cancel{N!}^N \cancel{n!} \cancel{(N-n-1)!}} = \frac{(N-n)}{N}$$

*proof*

The probability of  $i = 1$  in samples is,

$$1 - \frac{(N - n)}{N} = \frac{n}{N}$$

By induction it's said to be true for all unit.

## 4.2 Systematic Sampling

Divides the data in  $n$  (sample size) groups, order of each unit is maintained, comprising  $m$  (units). Each units can be selected with probability  $1/m$  within each group. Note that, I assume this is the rule to avoid bad sampling, “systematic sampling provides only  $m$  possible sets of samples”,

Using the earlier example, therefore I only have two sample  $S_1 = \{i, j, k\}$  and  $S_2 = \{x, y, z\}$  since  $6/3 = 2$  (units for each group). Note that, similar to SRS every unit in the population has the same probability. Furthermore, each unit is selected with probability  $1/m$  and each sample (combination) can be selected with probability  $1/m$  as well. I assume that they share the property.

The advantage is less complexity than SRS. SysS is used much in manufacturing in relation to a time period. For example, checking quality of product every 30 minutes, to have SS just pick randomly any time in the 30-minute interval. SysS also lead to less mistakes and improper short cuts in sampling that would lead to a haphazard or convenience sample. A clear disadvantage is that “the period for systematic sampling may coincide with particular patterns in the process or population”. In other words, the sample size needs to be a multiple of the population.

## 4.3 Stratified Sampling

SRS and SysS do not account for the implicit characteristics dividing the population into subpopulations, for example gender, country, etc. divide people. When the size of the subpopulation are substantially different then not all type of units is collected. For examples, country A has 10,000,000 females but just 100 males then there is a high chance that no units of males are collected. However, SRS and SysS assume the unit aligns with the time order that means, say, a product in a factory is produced at time  $t$ , the next one is at time  $t + 1$  and so on.

StrS accomodates the problem by setting a sample size of each subpopulation to a certain percentage of the number of units of the subpopulation. Like a fair share.

StrS works with time periods as well. By that, The population divided into  $n$  groups, whose orders are maintained, comprising  $m$  units with selecting probability  $1/m$ . The number of samples is the following,

$$K = \frac{N_1!}{n_1!(N_1 - n_1)!} \times \frac{N_2!}{n_2!(N_2 - n_2)!} \times \cdots \times \frac{N_M!}{n_M!(N_M - n_M)!} \quad (\text{II.2})$$

where  $N_i$  is the population size of each stratum  $i$ ,  $n_i$  is the sample size from each stratum  $i$ , given that  $i \in \{1, 2, \dots, M\}$ ,  $M$  being number of strata. Note that  $\sum^M N_i = N$  and  $\sum^M n_i = n$ . For instance of the time periods of the 6-sized population and 3-sized sample,  $N_i = 2$ ;  $n_i = 1$ ;  $M = 3$  therefore the possible samples:  $2^3 = 8$  – Different from SysS.

Similarly to SRS and SysS, the probability of each sample is  $p_s = 1/K$ . The probability of any unit in stratum  $i$  is  $n_i/N_i$ . “If the ratio of  $n_i/N_i$  is the same across all strata, each unit in the population has the same probability of being collected in the sample. This type of stratified sampling is called *proportional stratified sampling*”.

## 4.4 My Assumption

The reason for the difference between StrS and SysS is because SysS only allows a single pick for all group whereas StrS not only allow multiple sample from a group but also multiple parsing through

all group. StrS sort of like the Sequential Sampling from SRS. That is why the number of sample for SysS is,

$$K = \frac{m!}{1!(m-1)!} = m \quad (\text{II.3})$$

## 4.5 Cluster Sampling

Sometimes when direct sampling is not possible. Say, the lack of housing data but can be resolved by looking at images of region, groups or clusters of houses. Therefore, *cluster sampling* involves randomly selecting *clusters of units*. However, cluster sampling create groups of characteristics, which need to be acknowledged and analysed carefully to make proper statement about the whole population, because of its hierarchical structure. For examples, the metropolitan regions are densely packed with accomodations, but otherwise for rural areas.

Cluster sampling can be performed as *single-stage* or *multiple stages*. For single-stage, a random clusters are sampled and all unit from there is chosen. For multiple stages, say 2-stage, first random clusters are sample, then random units from such clusters are collected. The  $m$ -stage is the  $m$  depth of the sample procedures. The first sample units of the first stage is the *primary cluster units*.

The example of the sampling process is in page 52 of the book. Not written here yet since do not understand the probability of 1/6.

## 5 Evaluating Estimators Given Different Sampling Plans

After a plentiful of basics sampling procedures, to evaluate their “closeness” to the population’s theoretical value, it’s important to understand them through *bias*, *mean squared error* (MSE), and *standard error* (SE) to measure the closeness to know the performance of the estimators.

### 5.1 Generic Formulation of Sampling Plans

#### Definition

Let  $n$  be the sample size, and the population of units be  $\Omega = \{1, 2, \dots, N\}$ , where  $N \geq n$ . Let  $S_k \subset \Omega$  for  $k = 1, 2, \dots, K$  (samples) s.t  $S_k$  has  $n$  unique units from  $\Omega$  and  $\cup_{k=1}^K S_k = \Omega$ . Each  $S_k$  has a corresponding probability  $\pi_k > 0$  s.t  $\sum_{k=1}^K \pi_k = 1$ . Let  $S_k$  be unique, that is,  $S_k \neq S_l$  when  $k \neq l$ . It is not required for  $S_k \cap S_l = \emptyset$ . A sampling plan is

$$\mathcal{P} = \{(S_1, \pi_1), (S_2, \pi_2), \dots, (S_K, \pi_K)\} \quad (\text{II.4})$$

In other words, a set of samples and their corresponding probability.

Note that all of the previous random samplings obey this definition. The sampling plan results in set of probabilities  $p_1, p_2, \dots, p_N$  for units  $1, 2, \dots, N$  with  $p_i > 0$ , that is, the relation  $\hat{\mathcal{P}} = \{(p, \omega) \mid p \in \mathbb{P} \text{ and } \omega \in \Omega\}$ . It is not necessary that  $\sum_{i=1}^N p_i = 1$ . Recall those previous examples, the 6-pop and 3-sample one, we know that each sample has some probability, and in each sample the unit appear in some amount of time thus, somehow, from there the probability of the unit can be estimated.

The sampling plan has all the *necessary* information to analyse *the quality of the sampling procedure*. Even if with different sampling techniques, the quality solely relies on  $S_k$ ’s and  $\pi_k$ ’s.

### 5.2 Bias, Standard Error, and Mean Squared Error

#### 5.2.1 Notations

Consider a population of  $N$  units, and a *single* characteristic of the population is consider. Then, each unit  $i$  has a single theoretical value  $x_i$ . The population’s parameter of interest is defined by

$\theta := \theta(\mathbf{x})$ , where  $\mathbf{x} = \{x_1, x_2, \dots, x_N\}$  as some sort of calculation on all theoretical value, for example, population mean, or variance.

The sample  $S_k$  of size  $n$  is a set  $S_k = \{i_1, i_2, \dots, i_n\}$  where  $i_h = \{1, 2, \dots, N\}$  s.t  $i_h \neq i_l$  when  $h \neq l$ . Each sample  $S_k$  has an observation row-vector  $\mathbf{x}_k^\top = (x_{i_1}, x_{i_2}, \dots, x_{i_n})$ . By definition, the vector is written in vertical column, so we need to have a tranpose. The *descriptive statistics* can be computed by  $\hat{\theta}_k = T(\mathbf{x}_k)$ , an *estitmate* for the population parameter  $\theta$ . Most of time  $T = \theta$ , but different  $T$  is used depening on *sampling plan*. Note that, it is the **function**  $T$  and the **calculation**  $\theta$ , those mean, variance, etc. formulae.  $T$  is an *estimator*.

Generally,  $\hat{\theta}_k$  is an estimate of the population param. A good estimate is when sample is random and sample result is quite close to the pop. param.

**Weighted Average** One of  $T$  functions can be the weighted average divided by the sum of all weights ([Math Stack Exchange](#)), given by

$$\sum_{i \in S_k} \frac{w_i x_i}{1} \quad \text{s.t} \quad \sum_{i \in S_k} w_i = 1 \quad (\text{II.5})$$

### 5.2.2 Bias

The bias of an estimator  $T$ , given a sampling plan for parameter  $\theta$  is defined as

$$\text{bias} = \sum_{k=1}^K \hat{\theta}_k \pi_k - \theta = \mathbb{E}(T) - \theta \quad (\text{II.6})$$

If the bias is zero this means through repetitively taking samples from the sampling plan to calculate the statistics of interest, the average of all those statistics is equal to the true population parameter. Moreover, the estimator is said to be *unbiased*. Bias is not necessary bad.

**Expected Population Parameter** It is defined as

$$\mathbb{E}(T) = \sum_{k=1}^K \hat{\theta}_k \pi_i \quad (\text{II.7})$$

### 5.2.3 Mean Squared Error

A small bias only shows *on average* how close the sample's stats. are to the population parameter, but not individual stats. MSE concerns with the variability of the estimator accross different samples, defined as,

$$\text{MSE} = \sum_{k=1}^K \left( \hat{\theta}_k - \theta \right)^2 \pi_k \quad (\text{II.8})$$

The smaller MSE is the better the sampling plan. Sometimes the root mean square error (RMSE) is reported, which is simply  $\text{RMSE} = \sqrt{\text{MSE}}$

### 5.2.4 Standard Error

$$\text{SE} = \sqrt{\sum_{k=1}^K \left( \hat{\theta}_k - \mathbb{E}(T) \right)^2 \pi_k} \quad (\text{II.9})$$

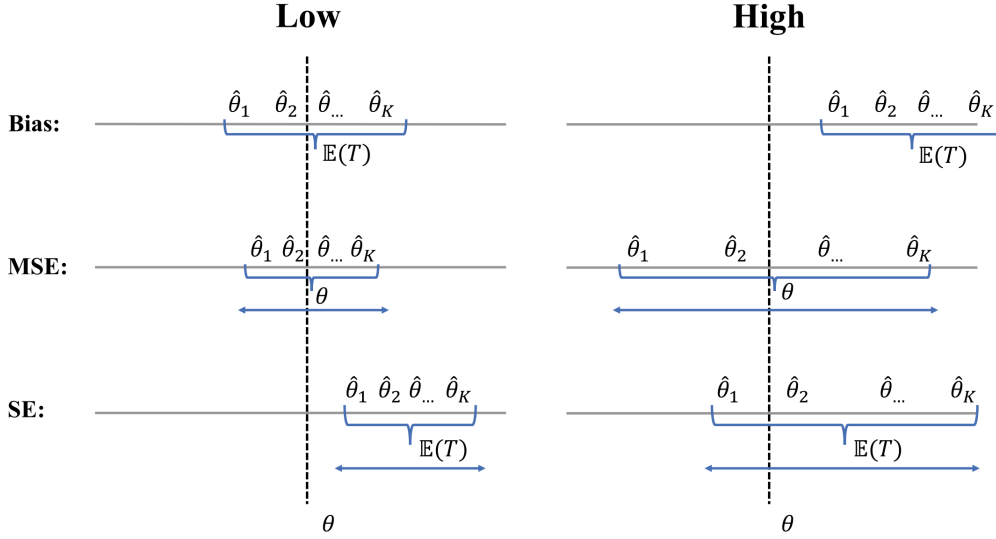
The standard error of an estimator is used as a measure to represent our uncertainty regarding an estimate. In many examples the standard error contains population parameters that we do not know. To use standard errors in practice we have to estimate the standard error as well.



**RMSE and SE** When the estimator is unbiased the RMSE and SE are identical, given by,

$$\text{RMSE} = \sqrt{\text{SE}^2 + (\mathbb{E}(T) - \theta)^2} \quad (\text{II.10})$$

Hence, the RMSE is never smaller than SE.



## 6 Estimation Of The Population Mean

When population mean  $\mu = \sum_{i=1}^N x_i/N$  is in the form of weighted average, the bias, MSE, and SE can be formulated mathematically when sampling plans are SRS, SysS, StrS, cluster sampling. Given some observations from the sample  $S_k$ , there occurs many times that some observations are more important than others. In practice, this could be due to gender, age, etc. Therefore, we use a weighted average, a weighted average of a sample  $S_k$  is,

$$\bar{x}_{w,k} = \sum_{i \in S_k} w_{ik} x_i \quad \text{s.t.} \quad \sum_{i \in S_k} w_{ik} = 1 \quad (\text{II.11})$$

Note that the weight depends on the sample  $S_k$ , but in this book assumes otherwise, that is,  $w_{ik} = w_i$ . This section concerns about the weighted average estimator for all the sampling plans.

### 6.1 Simple Random Sampling

The arithmetic mean in such sampling is the only unbiased estimator for the population mean in the class of weighted averages. Recall that, the expected population parameter is a weighted average of sampling plans.

$$\mathbb{E}(T) = \sum_{i=1}^K \hat{\theta}_k \pi_k$$

*proof*

To demonstrate that  $\bar{x}_k$  is unbiased, let recall from [Section 4.1](#), every value appear 10 out of 20 samples, so for a unit  $i$  it appear in the all  $K$  samples at most

$$K \frac{n}{N} \quad (\text{II.12})$$

From [Section 5.1](#),  $\cup_{k=1}^K S_k = \Omega$ . Instead of calculating the mean of all SRSs as individual unit in each sample, that is, (By Group then by

*proof*

element)

$$\mathbb{E}(T) = \sum_{k=1}^K \bar{x}_k \frac{1}{K} = \sum_{k=1}^K \sum_{i \in S_k} x_i \frac{1}{nK}. \quad (\text{II.13})$$

From Equation (II.12), the occurrence of each unit is known, the all the sample means can be calculated via that. Thus,

$$\sum_{k=1}^K \sum_{i \in S_k} x_i = \sum_{i=1}^N \frac{nK}{N} x_i \quad (\text{II.14})$$

Multiply  $\frac{1}{nK}$  to both side,

$$\mathbb{E}(T) = \frac{1}{N} \sum_{i=1}^N x_i = \mu$$

This makes the bias equal to

$$\text{bias} = \mathbb{E}(T) - \mu = 0. \quad (\text{II.15})$$

The MSE of the single arithmetic mean is,

$$\text{MSE}(\bar{x}_k) = \sigma^2 \left( \frac{N}{N-1} \right) \left( 1 - \frac{n}{N} \right) \frac{1}{n} \quad (\text{II.16})$$

where  $\sigma^2 = \sum_{i=1}^N (x_i - \mu)^2 / N$  is the population variance. When  $n = N$ , the MSE is not necessarily equal to 0. Given that  $\text{MSE} = \text{bias}^2 + \text{SE}^2$ , say now the arithmetic mean is not fully unbiased. For example,  $c\bar{x}_k$  then  $\text{bias} = (c-1)\mu$  and,

$$\text{SE}^2 = c^2 \sigma^2 \left( \frac{N}{N-1} \right) \left( \frac{1 - \frac{n}{N}}{n} \right) \quad (\text{II.17})$$

So when  $n = N$ ,

$$\text{MSE}(c\bar{x}_k) = [(c-1)\mu]^2 \quad (\text{II.18})$$

When  $n < N$ , the MSE of  $c\bar{x}_k$  would be lower than  $\bar{x}_k$  but this depends heavily on  $c$  and  $\mu$ .

### 6.1.1 Estimation of the MSE

Because MSE now depends on the population variance, we need to estimate it. The unbiased estimator for the population variance is

$$\hat{\sigma}^2 = \left( \frac{N-1}{N} \right) s_k^2 \quad (\text{II.19})$$

where  $s_k^2 = \sum_{i \in S_k} (x_i - \bar{x}_k)^2 / (n-1)$  is the sample variance. Therefore,

$$\hat{\text{MSE}}(\bar{x}_k) = \frac{(N-n)s_k^2}{Nn} \quad (\text{II.20})$$

## 6.2 Systematic Sampling

Assume that the population is perfectly splitted into  $n$  group of size  $m$ , each group select the  $k^{\text{th}}$  unit. The samples are  $S_1, S_2, \dots, S_m$  where  $S_k = \{k, k+m, k+2m, \dots, k+(n-1)m\}$ , each sample has the same probability  $1/m$ , the sample's arithmetic average is

$$\bar{x}_k = \frac{1}{n} \sum_{i=1}^n x_{k+(i-1)m} \quad (\text{II.21})$$

If the population is perfectly splitted, that is,  $N = mn$  the unbiased estimator of the population mean  $\mu$  is  $\bar{x}_k$ . Given that, (By element then by group)

$$\mu = \frac{1}{N} \sum_{h=1}^m \sum_{i=1}^n x_{h+(i-1)m} \quad (\text{II.22})$$

The MSE is

$$\sigma^2 = \frac{1}{N} \sum_{h=1}^m \sum_{i=1}^n (x_{h+(i-1)m} - \bar{x}_h)^2 \quad (\text{II.23})$$

And the population variance is,

$$\sigma^2 = \frac{1}{N} \sum_{k=1}^m \sum_{i=1}^n (x_{k+(i-1)m} - \mu)^2 \quad (\text{II.24})$$

The SysS is more efficient then SRS when the population variance is large.

### 6.2.1 Estimation of the MSE

In the general setting for systematic sampling, an unbiased estimation of the MSE is not possible. However, given that there is no systematic difference between units based on their position in the groups (and the ratio of sample and population size is an integer). By (Section 4.2), I assume that this mean both the probability of the sample and unit is the same, thus follow the property of SRS. Then, the MSE of the sample average under systematic sampling becomes equal to the MSE of the sample average under simple random sampling.

## 6.3 Stratified Sampling

### 6.3.1 Definitions

The section concerns with the properties of weighted average under StrS. To indicate unit  $i \in \{1, 2, \dots, N_h\}$  for units in stratum  $h \in \{1, 2, \dots, M\}$  with  $N_1 + N_2 + \dots + N_M = N$ , use the indices  $(h, i)$ , i.e  $x_{hi}$ . The population mean is

$$\mu = \frac{1}{N} \sum_{h=1}^M \sum_{i=1}^n x_{hi} = \sum_{h=1}^M w_h \mu_h \quad (\text{II.25})$$

where  $w_h = N_h/N$  and  $\mu_h = \sum_{i=1}^{N_h} x_{hi}/N_h$ . Therefore, the population mean is the weighted mean of strata mean  $\mu_h$ . The variance in stratum  $h$  is defined as

$$\sigma_h^2 = \frac{1}{N_h} \sum_{i=1}^{N_h} (x_{hi} - \mu_h)^2 \quad (\text{II.26})$$

The relationship between the population variance and the strata variances is given by

$$\sigma^2 = \frac{1}{N} \sum_{h=1}^M \sum_{i=1}^{N_h} (x_{hi} - \mu_h)^2 = \sum_{h=1}^M w_h \sigma_h^2 + \sum_{h=1}^M w_h (\mu_h - \mu)^2 \quad (\text{II.27})$$

The first part represents a weighted mean of the within strata variances and the second part represents a weighted mean of the squared distances of the strata means to the population mean. They may be referred to as within and between (strata) variances.

### 6.3.2 MSE of Sample Mean

Let the stratum sample size be  $n_h$ , where  $\sum_{h=1}^M n_h = n$ . The possible sample for the stratum  $h$  is  $S_{h,1}, S_{h,2}, \dots, S_{h,K_h}$  where

$$K_h = \frac{N_h!}{n_h!(N_h - n_h)!}$$

The sample mean of a sample  $k$  of stratum  $h$  is,

$$\bar{x}_{h,k} = \sum_{i \in S_{h,k}} \frac{x_{h,i}}{n_h} \quad (\text{II.28})$$

and the sample variance of such is,

$$s_{h,k}^2 = \sum_{i \in S_{h,k}} \frac{(x_{h,i} - \bar{x}_{h,k})^2}{n_h - 1} \quad (\text{II.29})$$

Since each stratum use SRS to sample then the sample mean  $\bar{x}_{h,k}$  is unbiased, moreover the bias and SE follows the theory of SRS. Thus, SE can be estimated with

$$\hat{\text{SE}}(\bar{x}_{h,k}) = \sqrt{\frac{1 - f_h}{n_h} s_{h,k}^2} \quad (\text{II.30})$$

where  $f_h = n_h/N_h$

The population mean can be estimated with,

$$\hat{\mu} = \bar{x}_k = \sum_{h=1}^M w_h \bar{x}_{h,k} \quad (\text{II.31})$$

given that  $w_1 = w_2 = \dots = w_M$ . The intuition is simply that each sample mean  $\bar{x}_{h,k}$  is unbiased, and plus add all the stratum create the population. To be unbiased, the weights need to be all similar. All because each stratum follows the theory of SRS.

Firstly the property of SE is that  $\text{SE}(cT) = c\text{SE}(T)$ . Secondly, using SRSs for different strata is not related, so  $\sum_{h=1}^M \text{SE}(w_h \bar{x}_{h,k}) = \text{SE}(\bar{x}_k)$ . Finally, the  $\text{SE}^2(\bar{x}_{h,k}) = \text{MSE}(\bar{x}_{h,k})$  since  $\bar{x}_{h,k}$  is unbiased. Therefore the MSE of  $\bar{x}_k$  is,

$$\begin{aligned} \text{MSE}(\bar{x}_k) &= \sum_{h=1}^M w_h^2 \text{SE}^2(\bar{x}_{h,k}), \quad \text{Using the first property, and the bias is 0 so only the squared SE.} \\ &= \sum_{h=1}^M w_h^2 \text{MSE}(\bar{x}_{h,k}) \\ \text{MSE}(\bar{x}_k) &= \sum_{h=1}^M w_h^2 \sigma_h^2 \frac{N_h(1 - f_h)}{(N_h - 1)n_h} \end{aligned} \quad (\text{II.32})$$

### 6.3.3 Estimation of MSE

The estimation can be formulated easily that is the  $\text{MSE}(\bar{x}_{h,k}) = \hat{\text{SE}}^2(\bar{x}_{h,k})$ , therefore

$$\hat{\text{MSE}}(\bar{x}_k) = \sum_{h=1}^M w_h^2 s_{h,k}^2 \frac{1 - f_h}{n_h} \quad (\text{II.33})$$

### 6.4.1 Single-stage Definition

For cluster sampling, perform a SRS of  $m$  clusters from total of  $M$  clusters. Given the cluster  $h$ , the sample mean is  $\bar{x}_h$  and its population of units is  $N_h$ . Since single-stage cluster take all of the units, hence  $\bar{x}_h = \mu_h$ . The population mean is defined as

$$\mu = \frac{\sum_{h=1}^M \sum_{i=1}^{N_h} x_{hi}}{\sum_{h=1}^M N_h} \quad (\text{II.34})$$

The use SRS's theory to estimate  $\mu$  is not the best choice in this case. Moreover, the arithmetic mean of cluster means is

$$\bar{x} = \frac{\sum_{h=1}^m \bar{x}_h}{m} \quad (\text{II.35})$$

which is an unbiased estimator of the given,

$$\sum_{h=1}^M \frac{\mu_h}{M} = \sum_{h=1}^M \sum_{i=1}^{N_h} \frac{x_{hi}}{M N_h} \quad (\text{II.36})$$

Firstly, in the first stage using SRS to sample the clusters, then the Equation (II.35) would be intuitively be the unbiased estimate of  $\hat{\mu}$ . However, if so, it is treating every cluster with the same size, and not accounts for cluster with difference in size. Which is why it is the unbiased estimator of only the Equation (II.36) where  $N_1 = N_2 = \dots = N_h$ .

### 6.4.2 Single-stage Estimation

Let the total sum of observations of the units in collected clusters, which is just using the mean of each cluster as the representative value for all of the values in that cluster, be

$$\sum_{h=1}^m N_h \bar{x}_h \quad (\text{II.37})$$

**Fixed-number Cluster's Size** When the cluster size is considered to be fixed then the estimator is

$$\bar{x}_F = \frac{\sum_{h=1}^m N_h \bar{x}_h}{\sum_{h=1}^m N_h} \quad (\text{II.38})$$

**Random-number Cluster's Size** When the cluster size is considered to be random then the estimator is

$$\bar{x}_R = \frac{\sum_{h=1}^m N_h \bar{x}_h}{N \frac{m}{M}} = \frac{M}{mN} \sum_{h=1}^m N_h \bar{x}_h \quad (\text{II.39})$$

where  $N(m/M)$  is the expected number of units in a single-stage cluster sample. Since each

Note that when  $N_1 = N_2 = \dots = N_h$  then  $\hat{\mu} = \bar{x} = \bar{x}_R = \bar{x}_F$ . Check pp. 70–71 for the example, I have yet to understood the calculation there. But it has shown that  $\bar{x}_R$  is an unbiased estimator of  $\mu$ .

**Comparison of Two Estimators** If the cluster means do not vary a lot, the estimator  $\bar{x}_F$  would be preferred over  $\bar{x}_R$ , but when there is relatively large variation across cluster means, then the preference is otherwise. In other words, when the cluster means are truly heterogeneous,  $\bar{x}_R$  provides the smallest standard error.

Intuitively, Equation (II.38) almost resemble the variance formula in Equation (I.6), this mean that  $\bar{x}_F$  is trying to average the total sum of observation by dividing with all the collect cluster size to address the problem of the Equation (II.35) (unweighted average). This let each of the cluster mean pull each other, where the large cluster will create stronger pull, as in higher weight, to land on the correct  $\hat{\mu}$ . Then, if the cluster means have high variability then  $\bar{x}_F$  suffers.

### 6.4.3 Estimation of The MSE

The estimator for the MSE of  $\bar{x}_R$  is

$$\hat{\text{MSE}}(\bar{x}_R) = \frac{M - m}{N^2 M m (m - 1)} \sum_{h=1}^m (M N_h \bar{x}_h - N \bar{x}_R)^2. \quad (\text{II.40})$$

Since  $\bar{x}_R$  is unbiased then  $\text{SE}(\bar{x}_R) = \sqrt{\hat{\text{MSE}}(\bar{x}_R)}$

For the two-stage cluster, fixed-number and random-number estimators are the same. The only difference is that  $\bar{x}_h \neq \mu_h$ , but  $\bar{x}_h = \hat{\mu}_h$  (an unbiased estimator), because each cluster is sampled using SRS. Therefore, the bias of the bias of  $\bar{x}_F$  and  $\bar{x}_R$  of two-stage and single-stage are equal. However, MSE of those two are different.

## 7 Estimation of the Population Proportion

Population proportion is same as population mean where the values are now binary, that is,  $z_1, z_2, \dots, z_N$  where  $z_i \in \{0, 1\}$ . The population proportion is obtained by

$$\eta = \sum_{i=1}^N \frac{z_i}{N} \quad (\text{II.41})$$

Binary variables are for unit that only has two outcomes. If a binary outcome is within a range say, over 18 is an adult and not if less than 18. The notation is  $z_i = 1_{[18, \infty)}(x_i)$ , with  $1_A(x)$  an indicator variable being equal to 1 if  $x \in A$  and 0 otherwise. The population proportion can be written as

$$\eta = \frac{1}{N} \sum_{i=1}^N 1_{[18, \infty)}(x_i)$$

The population variance  $\sigma^2$  for the binary values is

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (z_i - \eta)^2 = \sum_{i=1}^N \frac{z_i^2}{N} - \eta^2 = \sum_{i=1}^N \frac{z_i}{N} - \eta^2 = \eta(1 - \eta) \quad (\text{II.42})$$

Thus the population mean and population variance for binary variables are perfectly related. Knowing either of them can calculate the others.

### 7.0.1 Sample Average and Sample Variance

The sample average is

$$\hat{\eta}_k = \sum_{i \in S_k} \frac{z_i}{n} \quad (\text{II.43})$$

The sample variance is

$$s_k^2 = \sum_{i \in S_k} \frac{(z_i - \hat{\eta}_k)^2}{n - 1} = \frac{n \hat{\eta}_k (1 - \hat{\eta}_k)}{n - 1} \quad (\text{II.44})$$

Since population proportion is seen as population mean the theories from [Section 6](#) can be applied to population proportion. Practically, just replace the variable  $x$  with the variable  $z$ .

## 8 Estimation of the Population Variance

### 8.1 Background

This section concerns with only the population variance of SRS, as for other sampling procedures are substantially harder. The population variance is defined as

$$\sigma^2 = \sum_{i=1}^N \frac{(x_i - \mu)^2}{N}$$

and if a sample  $S_k$  is collected using SRS of  $n$  units, the sample variance is defined as

$$s_k^2 = \sum_{i \in S_k} \frac{(x_i - \bar{x}_k)^2}{n - 1}$$

The sample variance is not an unbiased estimator of  $\sigma^2$ . As it estimates

$$\hat{\sigma}^2 = \frac{N\sigma^2}{N - 1} = \sum_{i=1}^N \frac{(x_i - \mu)^2}{N - 1}$$

I assume the reason it can estimate that is because SRS has an unbiased population mean estimator  $\bar{x}_k$ , and also the subtraction of 1 unit. However, when the population size is large the estimate is very close to the population variance. But to an unbiased all-case estimator of  $\sigma^2$  is

$$\hat{\sigma} = \frac{(N - 1)s_k^2}{N} \quad (\text{II.45})$$

Now, say if the sample variance use  $n$  instead of  $n - 1$ , the estimator, from [Equation \(II.45\)](#), would be

$$(N - 1) \sum_{i \in S_k} \frac{(x_i - \bar{x}_k)^2}{Nn} \quad (\text{II.46})$$

Then the expected value and bias consecutively

$$\frac{\frac{(n - 1)\sigma^2}{n} - \sigma^2}{n} \quad (\text{II.47})$$

If the ratio of the population variance and the sample is large the estimator suffers, but if only the sample is increasingly large then there would no long be a strong argument for using  $n - 1$ .

An intuitive reason for the use of  $n - 1$  in the sample variance is that we first have to estimate the population mean  $\mu$  with the sample average  $\bar{x}_k$  and therefore we loose “one degree of freedom” from all  $n$  observations or available degrees of freedom. The use of the sample average makes one observation redundant. The divisor  $n - 1$  is referred to as the number of *degrees of freedom* used to calculate the sample variance.

## 8.2 MSE

The MSE of the estimator  $\frac{Ns_k^2}{N - 1} = \sum_{i=1}^N \frac{(x_i - \mu)^2}{N - 1}$

$$\text{MSE}\left(\frac{Ns_k^2}{N - 1}\right) = \sigma^4 \left( \frac{(N - n)(N - 1)(Nn - N - n - 1)}{N(N - 2)(N - 3)n(n - 1)} \gamma_2 + \frac{2(N - n)}{(N - 2)(n - 1)} \right) \quad (\text{II.48})$$

where  $\gamma_2$  is the population excess kurtosis defined  $\gamma_2 = \sum_{i=1}^N \frac{(x_i - \mu)^4}{N\sigma^4} - 3$ . When the excess kurtosis is close to zero and the population is large with respect to the sample size, the MSE of the estimator  $(N - 1)s_k^2/N$  is close to  $2\sigma^4/(n - 1)$ . More generally, when the population size is large, the MSE is approximately  $\sigma^4(\gamma_2/n + 2/(n - 1))$ , which is often used as an approximation to describe the above MSE.

## 8.3 Estimation of The MSE

The MSE in [Equation \(II.48\)](#) can be estimated by substituting estimates for the population parameters  $\sigma^4$  and  $\gamma_2$ . An obvious and common estimator of the squared variance  $\sigma^4$  is of course  $(N - 1)^2 s_k^4 / N^2$ , even though this estimator is biased. An estimator for the population kurtosis  $\gamma_2$  was provided by  $g_2$  in

Equation (I.11). This estimator is unfortunately biased for estimation of the population kurtosis, but it is considered an appropriate estimator for the kurtosis. Using these two estimators into and taking the square root Equation (II.48), an estimator of the precision of the variance estimator  $(N-1)s_k^2/N$  is obtained. Thus an estimator of the MSE in Equation (II.48) is now

$$\frac{(N-1)^2 s_k^4}{N^2} \left( \frac{(N-n)(N-1)(Nn-N-n-1)}{N(N-2)(N-3)n(n-1)} g_2 + \frac{2(N-n)}{(N-2)(n-1)} \right) \quad (\text{II.49})$$

and in case we deal with large populations we may use the simplified version  $s_k^4[g_2/n + 2/(n-1)]$ .



## Chapter III

# Application Topics of Chapter 1 and 2

### 1 Assignment 2.1 In The Book

---

#### 1.1 Part 4

---

```
1 y <- choose(n, m)
```

---

For the question 4, it introduce the function choose which is just a counting rule using combination.

---

#### 1.2 Part 5

In the answer of Systematic Sampling, that is

---

```
1 group_1 <- c(1, 2, 3)
2 group_2 <- c(4, 5, 6)
3 group_3 <- c(7, 8, 9)
4 sample_2 <- rep(NA, 3)
5 sample_2[1] <- sample(group_1, size=1)
6 sample_2[2] <- sample_2[1] + 3
7 sample_2[3] <- sample_2[1] + 6 # or sample_2[2] + 3
8 sample_2
```

---

Output

---

```
1 [1] 2 6 7
```

---

Which is tedious to list all the group out. So there are better method. Firstly is using `split()` with `ceiling()`

---

```
1 N <- 9
2 n <- 3
3 m <- N/n
4 population <- c(1:N)
5 groups <- split(population, ceiling(population/m))
6 start <- sample(groups[[1]], size=1)
7 S_2 <- start + (0:(n-1)) * m
```

---

Or secondly is using `split()`, with `rep()`

---

```

1 groups <- split(population, rep(c(1:m), each=m))
2 start <- sample(groups[[1]], size=1)
3 S_2 <- start + (0:(n-1)) * m

```

---

### 1.3 Part 6

---

In the answer for the Stratified sampling

---

```

1 sample_3 <- rep(NA, 3)
2 sample_3[1] <- sample(group_1, size=1)
3 sample_3[2] <- sample(group_2, size=1)
4 sample_3[3] <- sample(group_3, size=1)
5 sample_3
6
7 [1] 2 6 7

```

---

Which is tedious to sample each of them on by one, therefore I can use `sapply()`. Note that because `split()` return a list with names, aka the group names. The `sapply()` will return a named vector, with name of the position on top each value, to remove that just use `unname()`.

---

```

1 N <- 9
2 n <- 3
3 m <- N / n
4 population <- 1:N
5
6 groups <- split(population, rep(1:n, each = m))
7
8 dat_3 <- sapply(groups, function(g) sample(g, size = 1))
9 unname(dat_3)

```

---

## 2 Exercise 2.2

### 2.1 Part 1

---

Note that using `as.data.frame()` or `data.frame()` directly result the same. Keep in mind that `cbind()` return matrix so if the vector contains different type it will covert all to character type, aka the string. Becareful using `cbind()`. The code shows to use `apply()`. The difference between `%in%` and `==` is that `%in%` is value matching and “returns a vector of the positions of (first) matches of its first argument in its second” This means you could compare vectors of different lengths to see if elements of one vector match at least one element in another. The length of output will be equal to the length of the vector being compared (the first one). On the other hand, `==` is logical operator meant to compare if two things are exactly equal. If the vectors are of equal length, elements will be compared element-wise. If not, vectors will be recycled. The length of output will be equal to the length of the longer vector. ([StackOverflow](#)). For example,

---

```

1 1:2 %in% rep(1:2,5)
2 #[1] TRUE TRUE
3
4 rep(1:2,5) %in% 1:2
5 #[1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE

```

---

```

6
7 1:2 == rep(1:2,5)
8 #[1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
9
10 rep(1:2,5) == 1:2
11 #[1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE

```

---

```

1 school <- c(1:6)
2 students <- c(59, 28, 90, 44, 36, 57)
3 notimmunised <- c(4,5,3,3,7,8)
4
5 dat <- data.frame(cbind(school, students, notimmunised))
6 dat <- transform(dat, proportion=round(notimmunised/students,digits = 3))
7
8 theta <- 30/314
9
10 Samples <- combn(dat$school, m=3, simplify=TRUE)
11
12 proportion_sums <- apply(Samples, MARGIN = 2, function(combo) {
13   sum(dat$notimmunised[dat$school %in% combo])/sum(dat$students[dat$school %in%
14   ↪  combo])
15 })
16 # proportion_sums <- apply(Samples, MARGIN = 2, function(combo) {
17 #   # mean(dat$proportion[dat$school %in% combo])
18 # })
19
20 bias <- `*`(sum(proportion_sums),1/20) - theta
21 bias
22 MSE <- `*`((sum((proportion_sums-theta)^2),1/20)
23 MSE #$$

```

---

## 2.2 Part 2

---

```

1 school <- c(1:6)
2 students <- c(59, 28, 90, 44, 36, 57)
3 notimmunised <- c(4,5,3,3,7,8)
4
5 dat <- data.frame(cbind(school, students, notimmunised))
6 dat <- transform(dat, proportion=round(notimmunised/students,digits = 3))
7 theta <- 30/314
8
9 group_1<-c(1,3)
10 group_2<-c(4,6)
11 group_3<-c(2,5)
12
13 samples <- choose(2,1)**3
14
15 all_samples <- t(unname(expand.grid(group_1, group_2, group_3)))
16 all_samples
17

```

```

18 proportion_sums <- apply(all_samples, MARGIN=2, function(combo){
19     sum(dat$notimmunised[dat$school %in% combo])/sum(dat$students[dat$school %in%
    ↪  combo])
20 })
21
22 bias <- `*`(sum(proportion_sums),1/8) - theta
23 bias

```

---

### 3 Some Notes for Chapter 2

Firstly the randomness is actually lying inside the sampling not in the population, as the population values are  $x_1, x_2, \dots, x_N$ , which as of now are constants. So how samples are drawn is random which is discussed in the chapter. This means that the estimator  $T$ , or the arithmetic mean  $\bar{x}_k$  which mostly covered in chapter 2 is random.

Let's link back to the sampling plans, that is,  $\{(S_k, \pi_k) \mid 1 \leq k \leq K\}$  where  $K$  is the number of samples. Each  $S_k$  is random, and goes with it a probability  $\pi_k$  (the probability of such plan being chosen). Therefore, the sampling plan creates a probability distribution of the samples, where the  $x$ -axis is the sample (object), the  $y$ -axis is  $\pi_k$ . It is not much interested for now since every sample is uniformly distributed, however when applying the estimator  $\bar{x}_k$  giving  $\hat{\theta}_k$  hence with the corresponding  $\pi_k$ , which may form a different distribution since some of the  $\hat{\theta}_k$  maybe share the same value. That is why from [Section 5.1](#) suffice it to say that the “quality of the sampling procedure relies solely on  $S_k$ 's and  $\pi_k$ 's”. Because when I have the distribution of  $\hat{\theta}_k$ , I can evaluate its closeness to the population parameter (population mean)  $\theta$  using MSE, bias, and SE. And  $\hat{\theta}_k$  relies on  $S_k$  via  $T$ . Overall it can be viewed as

$$\underbrace{\{(S_k, \pi_k)\}}_{\text{the plan}} \xrightarrow{T} \underbrace{\text{distribution of } \hat{\theta}_k}_{\text{some curve}} \longrightarrow \text{bias, SE, MSE.} \quad (\text{III.1})$$

# Chapter IV

## Probability Theory

### 1 Definitions of Probability

Firstly, *event* is defined as something that happens or it's seen as a result of something. For example, people around the world receiving EMJD's scholarship are considered events. So then, the probability of such events can be the ratio of the frequency of all event and the frequency of the opportunities for the event to occur. This means that all people that received the scholarship as a ratio of all people that submitted for the scholarship of that period.

The possible opportunities for an event to occur is a *population of units* and the events can be seen as the population units with a specific characteristic. The probability for an event  $A$  for a finite population can be given by

$$P(A) = \frac{N_A}{N} \quad (\text{IV.1})$$

where  $N_A$  is the number of units with characteristic  $A$  and  $N$  the size of the population.

The equation [Equation \(IV.1\)](#) is only correct only if each opportunity for the event to occur is as likely to produce the event as any other opportunity. But realistically, it's not the case, to receive a scholarship a people with higher academic achievement would have higher chance those having worse academic performance. However, the population of units can be divided into smaller subset, but this would create some very small subset and result in probability of either 1 or 0. Moreover, this is only for a finite population, but rolling a die until getting a 6 can be an infinite long sequence of actions.

Thus another better approach is getting the probability in a more theoretical way by having a proportion of the occurrence of an event obtained from infinitely many repeated and identical trials or experiments under similar conditions. For example, let  $A$  be the event that the die show 1. The  $P(A)$  can be *approximated* by

$$P(A) = \frac{n_A}{n} \quad (\text{IV.2})$$

where  $n_A$  is the number of throws that results in 1,  $n$  is the total number of throws. When the number of identical trials  $n$  is increased  $n_A/n$  converges to  $p$  which would be 1/6 if a die is fair. Therefore, an alternative definition is

$$P(A) = \lim_{n \rightarrow \infty} \frac{n_A}{n} = p \quad (\text{IV.3})$$

Clearly, this definition is only appropriate when repeated and identical trials can be conducted under almost similar conditions (e.g., gaming and gambling). Thus, [Equation \(IV.3\)](#) is more conceptual for real situations, since it is impossible to conduct infinitely many or even many of these trials in practice.

### 2 Probability Axioms

#### Definition

The complement of  $A$  is  $A^c$ . The occurrence of two events  $A$  and  $B$  at the same time is  $A \cap B$ , that is, *joint* or *mutual* event. The event of either  $A$  or  $B$  is  $A \cup B$ .

The probability of no event must be 0. Not having event is  $\emptyset$  hence  $P(\emptyset) = 0$ .  
A two *mutually exclusive* events are  $A \cap B = \emptyset$  hence  $P(A \cap B = \emptyset) = 0$ .  
The probability that event  $A$  occur is one minus the probability that event  $A$  doesn't occur. This is built upon the assumption that either  $A$  or  $A^c$  occur, that is,  $P(A \cup A^c) = 1$ . Overall,  $P(A) = 1 - P(A^c)$ .  
Two events are independent iff  $P(A \cap B) = P(A)P(B)$ , and it's denoted  $A \perp B$ .  
Note that any event with no-event is independent. If two events are positive and mutually exclusive then it mustn't be independent.

**Calculation Rules** Using the axioms, some calculations can be derived.

1. If  $A \perp B$  then  $A \perp B^c$ ,  $A^c \perp B$ , and  $A^c \perp B^c$ .
2.  $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ .
3. The law of total probability:  $P(A) = P(A \cap B) + P(A \cap B^c)$ .

### 3 Conditional Probability

In some situations, probability is about a subset of the whole population. Say what is the probability of congenital anomaly given that the baby is a boy. This is different from the joint events probability, because now the probability of the space of interest excluded type girl. Let event  $A$  represents a congenital anomaly and  $B$  represents the event of a male newborn baby. The conditional probability is denoted by  $P(A | B)$  given by

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad \text{only if } P(B) > 0 \quad (\text{IV.4})$$

Dealing with joint events  $A \cap B$ , it can be summarised by a contingency table.

|       | $A$                               | $A^c$                                 |          |
|-------|-----------------------------------|---------------------------------------|----------|
| $B$   | $P(A \cap B) = P(A B) P(B)$       | $P(A^c \cap B) = P(A^c B) P(B)$       | $P(B)$   |
| $B^c$ | $P(A \cap B^c) = P(A B^c) P(B^c)$ | $P(A^c \cap B^c) = P(A^c B^c) P(B^c)$ | $P(B^c)$ |
|       | $P(A)$                            | $P(A^c)$                              | 1        |

Table IV.1: The  $2 \times 2$  contingency table.

For each, the two probabilities of column  $A$  and  $A^c$  add up to  $P(B)$  of the event of that row, because

$$P(A | B) + P(A^c | B) = 1 \quad (\text{IV.5})$$

### 4 Measures of Risk

Conditional probabilities also play a dominant role in the investigation of *associations* between events. Two events or variables are considered associated when they are not independent. For example, probability of congenital anomalies maybe different given the gender of either boy or girl. If independence is not the case, a comparison of two conditional probabilities can be used to quantify the strength of the association. Different *association measures* or *measures of risk* exist for this purpose, such as *risk difference*, *relative risk*, and *odds ratio*. Firstly to see what is an *outcome* and what is an *explanatory variable*, use  $D$  to denote the event of interest (outcome, results, and disease).  $E$  to denote the event that may affect the outcome (risk factor, explanatory event, exposure). For example, lung cancer  $D$ , with smoking ( $E$ ) or not smoking ( $E^c$ ).

## 4.1 Risk Difference

---

The risk difference or excess risk is an absolute measure of risk

$$ER = P(D | E) - P(D | E^c) \quad (IV.6)$$

and  $ER \in [-1, 1]$ . If  $ER > 0$ , there is a greater risk of the outcome when exposed than when unexposed. If  $ER < 0$ , the exposure is a protective factor for the outcome. If  $ER = 0$ , the outcome is independent of the exposure. Intuitively, the ER measures how much  $E$  or  $E^c$  protect the population from the event  $D$ . For example,  $D$  is the disease and  $E$  is getting vaccinated. Let  $P(D | E) = n$  and  $P(D | E^c) = m$ , then if  $ER < 0$  this means that getting vaccinated protect the from getting the disease.

## 4.2 Relative Risk

---

The relative risk would compare the two conditional probabilities  $P(D | E)$  and  $P(D | E^c)$  by taking the ratio.

$$RR = \frac{P(D | E)}{P(D | E^c)} \quad (IV.7)$$

If  $RR > 1$ , the exposed group has a higher probability of the outcome than the unexposed one. If  $RR = 1$ , the outcome and exposure are independent. If  $RR < 1$ , the unexposed group has a higher probability of the outcome.

## 4.3 Odds Ratio

---

The odds ratio compares the odds for the exposed group with the odds for the unexposed group. The odds is a measure of how likely the outcome occurs with respect to not observing this outcome. Mathematically, it is

$$O = \frac{p}{1 - p} \quad (IV.8)$$

Therefore, the odd of exposed group is  $P(D | E)/(1 - P(D | E))$ , the odds of unexposed group  $P(D | E^c)/(1 - P(D | E^c))$ . The odd ratio is given by

$$OR = \frac{O_E}{O_{E^c}} = \frac{P(D | E)(1 - P(D | E^c))}{P(D | E^c)(1 - P(D | E))} = \frac{P(D^c | E)}{P(D^c | E^c)} \times RR \quad (IV.9)$$

If  $OR > 1$ , the exposed group has a higher probability of outcome. If  $OR = 1$ , the outcome is independent of the exposure. If  $OR < 1$ , the unexposed group has a higher probability of the outcome.

Note that the odds ratio and relative risk are always ordered, that is, either  $1 < RR < OR$  or  $OR < RR < 1$ .

|              |                                                                                                                                                                                                                                                                                           |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>proof</i> | <p>If <math>RR &gt; 1</math>, <math>P(D   E) &gt; P(D   E^c)</math>. Since <math>P(D^c   E) = 1 - P(D   E)</math>, and <math>P(D^c   E^c) = 1 - P(D   E^c)</math>, thus <math>P(D^c   E) &lt; P(D^c   E^c)</math>. Via <a href="#">Equation (IV.9)</a>, then <math>OR &gt; RR</math>.</p> |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 4.4 Example

---

Please check the p. 92 for the example. I have the count contingency table as the following,

Table IV.2:  $2 \times 2$  contingency table for Dupuytren disease and gender (counts)

| Exposure         | Disease outcome   |                        | Total |
|------------------|-------------------|------------------------|-------|
|                  | Dupuytren ( $D$ ) | No Dupuytren ( $D^c$ ) |       |
| Male ( $E$ )     | 92                | 256                    | 348   |
| Female ( $E^c$ ) | 77                | 338                    | 415   |
| Total            | 169               | 594                    | 763   |

Since the sample size  $n = 763$ . I can utilise the table to easily calculate the probability as the following

Table IV.3:  $2 \times 2$  contingency table for Dupuytren disease and gender

| Exposure         | Disease outcome    |                        | Total              |
|------------------|--------------------|------------------------|--------------------|
|                  | Dupuytren ( $D$ )  | No Dupuytren ( $D^c$ ) |                    |
| Male ( $E$ )     | $92/763 = 0.1206$  | $256/763 = 0.3355$     | $348/763 = 0.4561$ |
| Female ( $E^c$ ) | $77/763 = 0.1009$  | $338/763 = 0.4430$     | $415/763 = 0.5439$ |
| Total            | $169/763 = 0.2215$ | $594/763 = 0.7785$     | 1                  |

## 5 Sampling from Populations: Different Study Designs

The odd ratio is more complex and widely used than the relative risk. An important reason is because OR is symmetric in exposure  $E$  and outcome  $D$ , that is, although exchanging  $E$  and  $D$  position regarding the conditional probability, the OR is invariant. However, relative risk changes and so does ER. The symmetry of the odds ratio makes it possible to investigate the association between  $D$  and  $E$  irrespective of the way that the sample from the population was collected. There are particular ways to select sample from population in medical research: population-based (cross-sectional), exposure-based (cohort study), and disease-based (case-control study).

In cross-sectional study, firstly, all the counts will be filled in table as the enumeration of the sample goes, then calculate the probabilities later. For the cohort study and case-control study, the count of exposure and outcome is consecutively known therefore those values are filled before the sampling procedure occur. Then, probabilities are calculated as usual.

### 5.1 Cross-sectional Study

In a cross-sectional study, a SRS of size  $n$  is taken from the population. Then, for each unit in the sample, note that only a sample, the count of exposure  $E$  and outcome  $D$  is recorded. Those records will be summarised into 4 cells comprising  $(E, D)$ ,  $(E, D^c)$ ,  $(E^c, D)$ ,  $(E^c, D^c)$ . For the instance of a contingency table,

| Exposure         | Disease outcome   |                        | Total    |
|------------------|-------------------|------------------------|----------|
|                  | Dupuytren ( $D$ ) | No Dupuytren ( $D^c$ ) |          |
| Male ( $E$ )     | ...               | ...                    | $P(E)$   |
| Female ( $E^c$ ) | ...               | ...                    | $P(E^c)$ |
| Total            | $P(D)$            | $P(D^c)$               | 1        |

Table IV.4: Contingency Table Template

since it is using SRS from a population, the probabilities of  $P(E)$ ,  $P(E^c)$ ,  $P(D)$ , and  $P(D^c)$  are unknown.

The estimation of *population proportion* is unbiased, since SRS is used. For example,

Let a binary variable be  $z_i = 1_{E_i=1 \wedge D_i=1}(x_i)$ , where  $x_i$  is the observed values of unit  $i$ . The probability  $\hat{P}(E \cap D)$  would be the sample average of the binary variable, moreover, an



unbiased estimate of the population proportion  $P(E \cap D)$ . Given that

$$\hat{\eta}_k = \sum_{i \in S_k} \frac{z_i}{n} = \hat{P}(E \cap D)$$

From Table IV.1, all the other probabilities of each cell would be similar. For each of the probability in the table Table IV.1, MSE can be evaluate the those since they are simply sample mean of a binary variable. Since the  $2 \times 2$  contingency table with sample data provides proper estimates of the population proportions, the measures of risk that would use these estimates from the sampled contingency table are estimates of the population measures of risk.

## 5.2 Cohort Study

---

Cohort study uses the mechanism of stratified sampling. Each of the exposed and unexposed population is collected by performing SRS. Therefore, via Table IV.4, the probabilities of  $P(E)$ ,  $P(E^c)$  are known. This means, that the count of either  $P(E)$  or  $P(E^c)$  has been preselected before sampling and are fixed in the sample.

For example, the congenital anomalies with cohort study. Let  $D$  be having the anomaly and  $E$  for a boy. By selecting, 500 boys and 1000 girls. The probabilities in the contingency table is no longer estimating the one of the population. However, it gives a focus study on either exposed or unexposed group. Say, the  $P(E) \approx 0$  which means that the exposure is rare in the population, but it's known that  $P(D | E) = 1$ , that means when something is exposed it will have the outcome. So, in the population,  $P(D \cap E) \approx 0$  which is trivial. But in our sample, since we can predefine the size of the two groups, we can get  $P(D \cap E)_{\text{sample}} = n \gg 0$  to be some non-trivial number.

The joint probabilities in the contingency can't estimate the ones in the population. But, those risk measure can estimate the population's risk. The reason is that these measures use the conditional probabilities only, where conditioning is done on the exposure. The  $P(D | E)$  and  $P(D | E^c)$  in the sample do represent the conditional population probabilities.

## 5.3 Case-Control Study

---

Case-control study focus on the outcome group. It use the same mechanism from cohort study, but now the strate being the groups of outcome. The probabilities  $P(D)$  and  $P(D^c)$  are preselected before sampling. Similar to cohort study, the the probabilities from the contingency table cannot estimate the population probabilities.

The odd ratio can still be properly determined.

## 6 Simpson's Paradox